



PORTFOLIO

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I build mobile and web products, and I train computer-vision models to read medical images. I like problems that sit between disciplines — where code meets design, and design meets people who have to actually use the thing.

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A deep-learning study that detects pneumonia from chest X-ray images, comparing DenseNet-121 and ResNet-50 backbones with a Convolutional Block Attention Module (CBAM) bolted on to sharpen where the model looks.

HOW IT STARTED

Final-year undergraduate thesis, Universitas Negeri Semarang

MY ROLE

Individual project — I did the literature review, dataset preparation, model implementation, training runs, evaluation and the written thesis.

IMPACT

[TBD: isi hasil terukur Anda — mis. "CBAM lifted DenseNet-121 accuracy from 92.1% to 95.4%, with recall on the pneumonia class rising from 0.89 to 0.94"]. The attention maps also made the model's decisions visually inspectable, which matters in a domain where an unexplained prediction is not usable.

WHAT I LEARNED

That model architecture is the easy half. The harder half is being honest about your data — class imbalance, leakage between train and test splits, and the difference between a metric that looks good and a model a clinician could trust. I also learned to read papers as engineering documents rather than as finished truths.

STACK & TOOLS

Python

TensorFlow / Keras

Jupyter Notebook

DenseNet-121

ResNet-50

CBAM

GALLERY

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LINKS

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A mobile app built for PT Ngupoyo Rejeki Lestari Mulya that handles driver attendance and tracks the live position of each LPG delivery driver, so the operations team can see the fleet without phoning anyone.

HOW IT STARTED

Project-based internship, January – May 2025

MY ROLE

GROUP PROJECT — I worked as a mobile developer. My main responsibility was the front end: screen architecture, the attendance and tracking interfaces, and state handling. I also contributed on the backend side, wiring the API endpoints the app depended on.

IMPACT

The app replaced a manual attendance process and gave the operations team a live view of where every driver was during a delivery run. [TBD: tambahkan angka jika ada — mis. jumlah supir yang memakai aplikasi, berapa lama waktu rekap absensi terpangkas].

WHAT I LEARNED

Building for people who are outdoors, moving, and not interested in your app taught me more about interface design than any tutorial. Buttons had to be large, states had to be obvious, and the app had to behave sensibly when the signal dropped. I also learned how much smoother a team gets once the front end and backend agree on the data contract before either side starts writing.

STACK & TOOLS

React Native

Expo

TypeScript

Node.js

Tailwind CSS

GALLERY

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A travel-planning web app focused on Bali, where a user can assemble a day-by-day itinerary and arrange hotels, accommodation and transport in one place.

HOW IT STARTED

Team capstone project for the Web Development & UI/UX course at Celerates Academy (PT Mitra Talenta Group), Kampus Merdeka 2024

MY ROLE

GROUP PROJECT — I worked as a full-stack developer, building interface components on the front end and route handling on the Express side. During the same programme I was named The Most Active Student of the cohort.

IMPACT

Turned a scattered planning process — separate tabs for hotels, transport and schedule — into one flow. [TBD: tambahkan hasil jika ada — mis. nilai/penilaian proyek, jumlah pengguna uji coba, atau umpan balik mentor].

WHAT I LEARNED

This was where the fundamentals actually landed: how a request travels from a browser to an Express route and back, and why structuring data well early saves a rewrite later. It was also my first real experience of splitting work across a team and merging it without breaking each other's code.

STACK & TOOLS

HTML

CSS

JavaScript

Express.js

Node.js

GALLERY

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The visual campaign for Interface 2023 and PKMMTJ 2023, two Computer Science department events: trailer videos, an Instagram feed identity, and a full photo and video record of both events.

HOW IT STARTED

Volunteer committee role — Media & Communications (Kominfo) division, July – September 2023

MY ROLE

GROUP PROJECT — as a Kominfo member I produced the trailer videos for both events, designed the Instagram feed layout, managed the @osjurilkom account, and handled documentation while the events were running.

IMPACT

The trailers and feed became the first thing prospective participants saw, and gave both events a consistent look instead of a folder of mismatched posters. [TBD: tambahkan angka jika ada — mis. jumlah penonton trailer, kenaikan pengikut, atau jumlah pendaftar].

WHAT I LEARNED

How to work to a deadline that will not move. An event happens on its date whether the trailer is ready or not, so I learned to plan backwards from the ship date and to cut scope rather than quality. Running an account also taught me that consistency beats individual brilliance — a feed is judged as a whole.

STACK & TOOLS

Video editing

Instagram content design

Event documentation

GALLERY

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During UNNES' community service programme in Desa Bogem, I produced the village profile video and designed and delivered a cyberbullying-awareness class for the village's children.

HOW IT STARTED

University community service programme (KKN Giat 13 UNNES), 1 October – 30 November 2025

MY ROLE

GROUP PROJECT — I was on the Media team. I directed and edited the village profile video, designed the programme's Instagram content, documented every activity across the two months, and wrote and taught the cyberbullying material to the children of Desa Bogem.

IMPACT

The village gained a profile video it can use to introduce itself, and a group of children got a first, concrete vocabulary for what online harm looks like and what to do about it. [TBD: tambahkan angka jika ada — mis. jumlah anak yang hadir, jumlah penonton video].

WHAT I LEARNED

Explaining cyberbullying to a room of children forced me to strip an idea down to its bones — no jargon, no assumed context. It is the same skill as writing a good error message or naming a button clearly, and I have used it in every interface I have built since. I also learned that technology is only useful once you understand the people it lands on.

STACK & TOOLS

Video production

Instagram content design

Public speaking

Documentation

GALLERY

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